

ROHAN YARLAGADDA

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EDUCATION

Bachelor of Science in Engineering
Harvey Mudd College, Claremont, CA
Division III Varsity Tennis | CMS

May 2028

RELEVANT COURSEWORK

Material Science and Engineering, Advanced Systems Engineering, Introduction to Electrical Engineering, Digital Electronics and Computer Engineering, Applied Math for Engineers, Experimental Engineering

SKILLS

- Programming: Python, SystemVerilog, MATLAB, RISC-V Assembly, Java
- Software: LabVIEW, Solidworks, Quartus, QuestaSim, Excel
- Hardware and Bench Equipment: Altera FPGA, Oscilloscope Tracing, Breadboarding, Soldering, RLC Circuit Design

WORK EXPERIENCE

NSF Undergraduate Researcher in Semiconductor Materials, **Department of Materials Engineering**, Michigan State University, East Lansing, Michigan June 2026 - September 2026

- Extended SALSA, a GitHub-hosted materials-discovery pipeline, with automated convex-hull stability analysis to screen earth-abundant semiconductors for photocatalytic water-splitting
- Validated thermodynamic stability across four levels of theory (OPT2, SP, enthalpy, Gibbs free energy)
- Consolidated ~20 scripts into a unified Python suite for AFLOW/Materials Project scraping, CIF generation, CRYSTAL23 parsing, and hull plotting, cutting manual analysis from days to hours

Research Intern, Aravasc Inc., Sunnyvale, California

June 2024 - August 2024

- Engaged in experiments to develop a safe and reliable diagnostic for detecting HER2+ breast cancer
- Analyzed MSH2IA iron nanoparticles, polymers, and PEG amines using High-Pressure Liquid Chromatography (HPLC)

PROJECTS

Mechanical Engineer, Fluorescence-Guided Surgery Imaging System | PATH Imaging September 2026 - Present

- Building a low-cost, Raspberry Pi-based dual-camera system overlaying fluorescence imaging onto live visible-light video to guide surgeons
- Collaborating with a 7-person team and PATH Imaging liaison to deliver a solution at a fraction of commercial system costs
- Leading CAD for all camera mounting assembly, creating 3D-printed adapters to integrate C-mount lenses (16 mm and 25 mm) onto the Vision Components IMX412 RGB and TechNexion NIR fluorescence camera modules with precise alignment
- Developing firmware and image-processing code to capture, synchronize, and overlay fluorescence and visible-light video feeds in real time

Project Lead, Underwater Seafloor Mapping Robot | Harvey Mudd E79

January 2026 - June 2026

- Led a four person team to develop a low-cost seafloor mapping system within a \$50 budget, integrating laser triangulation and Raspberry Pi imaging to generate 3D topographic maps using MATLAB
- Designed and built temperature, pressure, and camera sensors allowing an underwater diving robot to collect sea data
- Utilized mechanical and electrical systems to optimize robot mobility and effectiveness
- Conducted multiple prototyping sessions and dry tests to ensure reliability and consistency

Project Lead, Hydrogel Compression Testing Device | Harvey Mudd E4

September 2025 - December 2025

- Formulated a low-cost, air-pressure-driven compression tester letting 4th and 5th graders measure hydrogel deformation, using popping boba as a food-safe test material for school outreach events
- Modeled full prototype in SolidWorks, then machined and fabricated components to combine off-the-shelf syringes, check valves, and PVC tubing into a modular, waterproof design at \$11.26/unit, under a \$15/unit budget

EXTRA CURRICULAR ACTIVITIES

Student-Athlete, CMS Men's Tennis, Claremont, California

August 2024 - Present

- NCAA Division III Varsity Tennis: Compete for CMS Men's Tennis, a top-5 nationally ranked team, balancing 20+ hours/week of training and travel with a rigorous engineering course load